Sanaul Haque

Apollo_Wang_Proposal

Course 608

Document Details

Submission ID
trn:oid:::3117:636163097

Submission Date
Oct 8, 2026, 10:35 PM GMT+6

Download Date
Oct 8, 2026, 10:37 PM GMT+6

File Name
Apollo_Wang_Proposal.pdf

File Size
166.3 KB

7 Pages

3,449 Words

18,834 Characters

13% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

20 Not Cited or Quoted 12%
Matches with neither in-text citation nor quotation marks

5 Missing Quotations 1%
Matches that are still very similar to source material

1 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

11% Internet sources
11% Publications
12% Submitted works (Student Papers)

Match Groups

- **20** Not Cited or Quoted 12%
  Matches with neither in-text citation nor quotation marks
- **5** Missing Quotations 1%
  Matches that are still very similar to source material
- **1** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

11% Internet sources
11% Publications
12% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	ucl.rl.talis.com 2%
2	Internet	di.ku.dk 1%
3	Internet	www.researchsquare.com 1%
4	Student papers	Tilburg University on 2024-06-20 1%
5	Publication	Anhar Sami Mohammed, Prajoy Podder, Maciej Zawodniok, Cihan Dagli. "Cognitiv... <1%
6	Publication	Zhang Chengjie, Yu Suiran. "Virtual Co-Embodiment Rehabilitation: An Innovative... <1%
7	Internet	arxiv.org <1%
8	Student papers	Georgia Institute of Technology Main Campus on 2025-11-24 <1%
9	Internet	downloads.hindawi.com <1%
10	Student papers	University of Strathclyde on 2025-10-27 <1%

11 Publication
Calvin C. Chandler, Ryan P. Serbin, James T. Frix, Marcie H. Siebert et al. "An intro... <1%

12 Student papers
Vrije Universiteit Brussel on 2026-08-28 <1%

13 Publication
Reginold John, Jose John, Joseph Mathew, Sebastian John. "Hybrid work and empl... <1%

14 Publication
Agnieszka Fudali-Czyż, Rafał Lewkowicz, Piotr Francuz, Paweł Stróżak et al. "An At... <1%

15 Publication
Stephen P.J. Goodman, Frank E. Marino. "Thirst perception exacerbates objective ... <1%

16 Internet
www.ijirset.com <1%

17 Student papers
Edge Hill University on 2026-03-25 <1%

18 Student papers
University of Essex on 2026-09-17 <1%

19 Internet
assets-eu.researchsquare.com <1%

20 Internet
journals.lww.com <1%

21 Internet
www.hindawi.com <1%

22 Publication
Barbara A. Wilson, Jill Winegardner, Caroline M. van Heugten, Tamara Ownswort... <1%

23 Student papers
Khalifa University of Science Technology and Research on 2023-12-07 <1%

24 Student papers
University of Warwick on 2026-06-29 <1%

Full Legal Name: Apollo Wang

Institution: Stuyvesant High School

Category: Junior

Class/Grade/Year: Class of 2028, 11th Grade

Country: United States of America

Submission Date: 9/20/2026

Registered Email Address: gorange2009@gmail.com

Research Topic: Autonomy support, Voluntary Persistence, Mental Fatigue, VR Rehabilitation.

Research Title: Autonomy support and voluntary persistence in a VR rehabilitation exercise under mental fatigue.

Introduction and background

Recovery of movement after stroke or traumatic brain injury depends on neuroplasticity, the brain's capacity to reorganize in response to stimulation. Neuroplasticity is dose-dependent [1]: it requires long-term, high-intensity, repetitive practice with the patient fully engaged. Sustained focus of that kind induces mental fatigue, the state following prolonged cognitive demand, which reduces motivation and concentration [2]. The limitation of the traditional model is therefore self-reinforcing: as fatigue accumulates, patients disengage from the repetition the exercises depend on, and the delivered dose falls below what plasticity requires. My training in competitive swimming led me to bring RPE (Rating of Perceived Exertion, a 0-10 scale of how hard an effort felt) [3] into this proposal. Studies of mentally fatigued athletes report a striking dissociation [4]: heart rate and blood lactate are unchanged at a given workload, yet the same work feels harder and performance declines. Mental fatigue acts on the perception of effort rather than on the muscle, and that should apply to anyone sustaining effortful work - including patients for whom prolonged repetitive exercise is precisely the prescription. If disengagement under fatigue reflects inflated perceived effort rather than physical incapacity, a motivational intervention is a legitimate lever. Of the proposed countermeasures [5] - nature exposure, personal task fit, transcranial direct current stimulation, and autonomy-supportive practice - I selected autonomy support: genuine choice rather than a fixed prescription. It requires no equipment or specialist operator, and self-determination theory [6] identifies autonomy as a basic psychological need that drives intrinsic motivation. Conventional therapy offers little to choose from, so I deliver it through virtual reality, where difficulty, scenery and target appearance are adjustable, and choice can be offered without altering the exercise itself. Reviews of VR for stroke rehabilitation have rated the underlying trials low or very low quality and called for studies of mechanism rather than further demonstrations of benefit [7].

The gap, research question and hypotheses

Previous work has shown benefits of VR-based rehabilitation [7, 8], characterised the effect of mental fatigue on athletic performance [4], and proposed countermeasures [5]. Fatigue after brain injury is not unstudied. Persistent fatigue is among the most commonly reported symptoms after stroke [9], and learner choice is not unstudied [10], since self-controlled practice is established in motor learning. What is missing is the intersection of these areas. Most research on perceived effort comes from athletics, with little attention given to rehabilitation adherence. Studies of self-controlled practice have also used rested participants and focused on skill acquisition, leaving persistence under fatigue largely unexplored. In VR research, researchers generally evaluate motivation across the system as a whole, and rarely manipulate individual components. This study excludes physiological measures because the expected effect should occur without changes in heart rate or lactate. It also excludes patients, whose participation would require clinical oversight beyond the scope of this initial study.

In mentally fatigued healthy adults performing an upper-limb VR reaching exercise, does being allowed to choose one's own session settings increase voluntary practice time beyond the required duration, compared with receiving those same settings without choosing them?

H1 (primary): with both groups mentally fatigued, participants given choice will voluntarily practice longer after the required session ends than participants given none. Outcome: additional practice time in seconds.

H2 (secondary): perceived exertion on the Borg CR10 scale (0-10) [3] will be lower in the

choice group than the control group, for objectively identical work.

H3 (manipulation check): subjective mental fatigue (VAS 0-100) will not be lower after the VR task than immediately before it, confirming fatigue was present throughout and that any difference in persistence cannot be explained by fatigue wearing off. This is a check rather than a hypothesis, since a non-significant difference is not evidence of no difference.

Methods

Part 1: Design and participants

Design: We will run a randomised study with two parallel groups: an experimental group and a yoked control group. Each person will take part in one session, followed by a safety check 24 hours later. We aim to recruit 40 participants using an online screening form shared at local high schools and university campuses. Participants will then be randomly assigned, with 20 in each group. The recruitment notice will state that taking part is voluntary and unpaid, is not linked to a course or grade, and can be declined without any penalty. Inclusion & Exclusion: We will recruit healthy adults aged 18–25. People will not be eligible if they have a diagnosed neurological injury or disorder, a history of seizures or photosensitive epilepsy, a vestibular disorder, severe motion sickness, or a concussion in the past six months. We will also exclude those with uncorrected visual impairment, an upper-limb injury that limits reaching, or medication use that affects alertness. Participants must have slept for at least five hours the night before the session. Because sleep, caffeine intake, and time of day can affect mental fatigue, all sessions will be held between 15:00 and 19:00. Participants will avoid caffeine for three hours before testing, and we will record their sleep and caffeine intake for use as covariates. Sample size: I plan to recruit 40 participants, with 20 in each group. With a two-sided alpha of 0.05 and 80% statistical power, this sample can detect an effect of about Cohen's $d = 0.91$, which is large. Detecting a moderate effect of $d = 0.50$ would take about 64 participants per group, which is beyond the scope of this study. If no significant difference is found, that will not show that smaller effects are absent. I will report the effect estimate and confidence interval so that this uncertainty is clear, rather than relying on the p-value alone. Before recruitment begins, I will create the group assignments in blocks of four, with separate sequences by sex. I will keep the assignments in sealed, opaque envelopes and open them only after consent.

Part 2: Inducing and verifying mental fatigue

Each participant first completes a Visual Analogue Scale (VAS), marking subjective mental fatigue 0-100 in answer to one question: how mentally fatigued do you feel right now? They then complete 30 minutes of a continuous incongruent Stroop task [11] - colour words such as RED and BLUE in mismatched ink, naming the ink colour while restraining the word - a standard fatigue induction. The VAS is then repeated to establish the pre-experiment baseline and confirm increased fatigue. During the Stroop task, the software saves response times and accuracy in five-minute intervals. Evidence that the induction worked will require both higher VAS scores and poorer Stroop performance. We will still include those who show little or no change, since excluding them after examining persistence results could bias the analysis. As an additional check, we will repeat the H1 analysis with each participant's change in VAS score entered as a pre-specified covariate.

Part 3: Main VR Task for each group

The groups are scheduled so they do not encounter one another, preventing either arm from

learning what the other received; with a single headset, participants run one at a time. Group A is the intervention group, Group B the control. Both complete the same seated 10-minute session of one upper-limb reaching exercise, in which three-dimensional targets appear at pseudo-random positions to be touched, at a distance scaled to each participant's arm length. Both groups begin with identical defaults: medium difficulty, a hospital-room environment, red targets against the background, and no music. Before the timer starts, Group A is told they may set three parameters - difficulty, environment and target colour and given 90 seconds at a settings menu; colour options are pre-restricted to maintain contrast, and every choice is logged. Participants in Group B do not see the settings menu. Each person receives the settings previously selected by their matched participant in Group A. The settings are presented as having been configured for them, without mentioning that another participant chose them. Before beginning the task, they spend 90 seconds looking around the scene. This keeps their headset time the same as that of Group A. This is the yoked design [10]: each control receives their partner's exact settings, so both arms experience an identical distribution of difficulties, environments and colours and differ only in who decided. Without yoking, the study is confounded, because Group A could simply lower the difficulty, practice longer, and report lower exertion for an easier task. At exactly 10:00 the participant is told the required practice is complete, that they have met it in full, and that they may stop or continue for up to a further 10 minutes. The primary outcome is seconds elapsed from this prompt until they stop or remove the headset, right-censored at 600 seconds. The Unity application records the time automatically, reducing recording errors and possible experimenter bias. The experimenter will leave the room while the participant decides whether to continue and will provide no encouragement.

Part 4: Full proposed procedure

Participants are screened for eligibility upon arrival. Eligible participants read and sign the consent form. They then complete the baseline SSQ [12] and note how much they slept the previous night and any recent caffeine intake. Before testing begins, I show them how to use the VR equipment and let them work through a short tutorial. I also go over the safety instructions at this point. Participants record their first VAS rating just before beginning the Stroop task. They will rate themselves again after the Stroop task and then select their settings for the VR activity. They then perform the task for 10 minutes and decide at the stop prompt whether they wish to continue. At the end of the session, participants complete the third VAS rating, the Borg CR10, and another SSQ before being debriefed. Participants complete a final SSQ online 24 hours later to screen for delayed simulator sickness. Participants are tested individually, with no more than four sessions per day. Each visit takes about 75 minutes. The SSQ helps identify participants who stop because they feel unwell.

Part 5: Analysis plan and predicted outcomes

The hypotheses, predictions, and statistical tests will be set before data collection begins. H1 is the primary hypothesis. We expect participants in Group A to continue practising for about one to six minutes longer after the stop prompt. Some participants may stop immediately, while others may continue until the time limit, so the persistence times are likely to be right-skewed and clustered at zero and at the cap. Given that shape, Mann-Whitney U (two-sided, alpha = 0.05) rather than a t-test, reported with rank-biserial correlation, a 95% confidence interval, medians and interquartile ranges, and the number reaching the cap. As a check on the main result, I will also analyse persistence using Kaplan–Meier curves and a log-rank test. Participants

who are still practising when they reach the time limit will be treated as right-censored. I will then run a sensitivity analysis that includes VAS change, sleep, and sex as covariates. H2 is a secondary hypothesis: I expect the choice group to report a lower average Borg CR10 score. This is the more theoretically notable prediction because, under yoking, the exercise is matched between arms, so a difference in perceived exertion shows that choice changed the experience of the task rather than the task. Test: Mann-Whitney U with rank-biserial effect size and confidence interval. H3, check. Prediction: fatigue remains at or above its post-induction level; 10 minutes of pleasant activity should not resolve 30 minutes of induced fatigue, and I do not predict that choice reduces fatigue - only that it changes what a fatigued person is willing to do. We will compare VAS scores at time points 2 and 3 using a Wilcoxon signed-rank test. If the result is not significant, we will not assume that fatigue stayed the same. We will also report the average change and its 95% confidence interval. We will set a 10-point difference before data collection as the smallest change we consider meaningful, and the confidence interval will show whether a change of that size remains plausible. We will report effect sizes and confidence intervals for every endpoint regardless of significance. We will show exclusions and dropouts in a flow diagram, and we will make de-identified data and code available.

Risks, Safety, and Ethics

My school research mentor, a qualified adult researcher, will review the protocol before submitting it to the relevant institutional and competition review bodies. I will not begin recruiting or screening participants until I receive written approval. My mentor will be present for every session, and two volunteers with current CPR and First Aid certification will also attend each testing day. All participants are adults and provide written informed consent covering the purpose, procedures, risks, data use, and the right to stop at any time without reason or consequence. It explicitly covers the induction. Participants are told the Stroop task is meant to make them mentally tired. Because participants come from campuses I am connected to, there is a real risk of implicit pressure to take part or to keep practicing to help my project. Participants will be recruited through an open advertisement, not approached personally, and no incentive will be offered. When they decide whether to continue with the task, the experimenter will leave the room so that they do not feel watched or pressured. Participants are identified by code number; the linking file is stored separately, encrypted, accessible only to me and my mentor, and destroyed after analysis. No video is recorded, and results are reported in aggregate. After the tutorial, participants are shown safety slides distinguishing normal fatigue responses (reduced motivation, difficulty concentrating) from abnormal symptoms (nausea, dizziness, eye strain), so they can recognize the latter themselves. Participants can stop the session at any point. If a participant looks distressed, I will stop the task. Anyone who feels unwell will stay seated with the spotter or me until they feel better. Participants remain seated throughout the VR activity, and we will keep a two-metre space around the chair clear. The VR program does not contain flashing images, and anyone with a history of photosensitive seizures will be screened out. I will wipe down the headset after every session and use a fresh face liner for each person.

Equipment, budget and approximate timeline

Requirements: one standalone VR headset (about $349); a school-provided Chromebook; hygiene supplies (about $30); printed consent and VAS, Borg CR10 and SSQ sheets (about $15); safety slides; and the Unity task under a free student license. Total about $394. Personnel are volunteers: an adult research mentor and two people holding current CPR and First Aid certification.

Weeks 1-3: complete the protocol, submit for institutional and competition review, begin the Unity build. Weeks 3-5: secure the mentor and first-aid volunteers; build the screening form, record instruction audio and safety slides; pilot with two or three volunteers to verify that setting choices, timing and continuation logging work. Weeks 5-6: approval received, recruitment opened. Weeks 6-13: data collection at up to four participants per day, about 32 testing days for 40 participants with margin for dropouts. Weeks 13-15: analysis, write-up, deposit of data and code.

Limitations

1.This study involves healthy volunteers, not patients. Young adults may differ from people with stroke or TBI in their usual level of fatigue, cognitive ability, and motivation. The findings can therefore tell us about the underlying mechanism, but they cannot establish a clinical benefit. 2. There is no rested comparison group. Since every participant completes the task after the fatigue induction, the study can show whether having a choice helps people persist when fatigued. It cannot show whether the effect is stronger under fatigue than under rested conditions. 3. The study measures persistence during a single session, not learning over time. It cannot tell us whether longer practice leads to better motor learning or clinical improvement. The sample size is also sufficient only for detecting large effects. If the study finds no difference, moderate effects may still be present. 4. The effect of choice may partly overlap with the novelty of using VR. Participants who are new to VR may spend more time exploring the available options simply because they are curious about them. 5. A single unblinded experimenter who holds the hypothesis, and two self-report measures susceptible to demand characteristics. Recorded instructions, concealed allocation, automatic timing, and the objective Stroop measure should help reduce these risks, although they cannot remove them entirely.

Interpretation and significance

If participants in the choice group practise for longer and rate the same amount of work as less effortful, the findings would suggest that choice supports persistence during mental fatigue by lowering the perceived cost of the task. The conclusion would be limited to this specific mechanism, not VR rehabilitation as a whole. Even so, it would address a gap identified in previous reviews by isolating one feature of the VR experience. It would also have a practical implication, since providing patients a choice would add little or no cost to rehabilitation. A different interpretation would be needed if most participants stop as soon as they are given the option. In that case, the session may have been too short, or the VR experience too unfamiliar, for an effect on persistence to emerge. Such a result would not show that autonomy support is ineffective. Whatever the outcome, the study will provide an initial estimate of the effect on persistence under fatigue and help determine the design and sample size of a follow-up study.

Future phases and my plans

If the present study finds a meaningful effect, the next step would be to test the approach across several sessions with patients who have neurological injuries. This would show whether giving patients a choice helps them complete more practice over time, which may support motor recovery. A patient study would need suitable exercises, clinical supervision, and full institutional review. In rehabilitation, an intervention can be effective only if patients are willing to complete it, so motivation should be part of its design.

References

[1] J. A. Kleim and T. A. Jones, "Principles of experience-dependent neural plasticity: Implications for rehabilitation after brain damage," *Journal of Speech, Language, and Hearing Research*, vol. 51, no. 1, pp. S225–S239, Feb. 2008, doi: 10.1044/1092-4388(2008/018).

[2] M. A. S. Boksem and M. Tops, "Mental fatigue: Costs and benefits," *Brain Research Reviews*, vol. 59, no. 1, pp. 125–139, Nov. 2008, doi: 10.1016/j.brainresrev.2008.07.001.

[3] G. A. Borg, "Psychophysical bases of perceived exertion," *Medicine and Science in Sports and Exercise*, vol. 14, no. 5, pp. 377–381, 1982.

[4] J. Van Cutsem, S. Marcora, K. De Pauw, S. Bailey, R. Meeusen, and B. Roelands, "The Effects of Mental Fatigue on Physical Performance: A Systematic Review," *Sports Medicine*, vol. 47, no. 8, pp. 1569–1588, Jan. 2017, doi: 10.1007/s40279-016-0672-0.

[5] H. Sun *et al.*, "Counteracting mental fatigue for athletes: A systematic review of the interventions," *BMC Psychology*, vol. 12, Art. no. 67, Feb. 2024, doi: 10.1186/s40359-023-01476-w.

[6] R. M. Ryan and E. L. Deci, "Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being," *American Psychologist*, vol. 55, no. 1, pp. 68–78, Jan. 2000, doi: 10.1037/0003-066X.55.1.68.

[7] A. Voinescu, J. Sui, and D. Stanton Fraser, "Virtual Reality in Neurorehabilitation: An Umbrella Review of Meta-Analyses," *Journal of Clinical Medicine*, vol. 10, no. 7, p. 1478, Apr. 2021, doi: 10.3390/jcm10071478.

[8] K. E. Laver *et al.*, "Virtual reality for stroke rehabilitation," *Cochrane Database of Systematic Reviews*, vol. 2025, no. 6, June 2025, doi: 10.1002/14651858.cd008349.pub5.

[9] T. B. Cumming, M. Packer, S. F. Kramer, and C. English, "The prevalence of fatigue after stroke: A systematic review and meta-analysis," *International Journal of Stroke*, vol. 11, no. 9, pp. 968–977, Oct. 2016, doi: 10.1177/1747493016669861.

[10] A. Lemos, G. Wulf, R. Lewthwaite, and S. Chiviacowsky, "Autonomy support enhances performance expectancies, positive affect, and motor learning," *Psychology of Sport and Exercise*, vol. 31, pp. 28–34, July 2017, doi: 10.1016/j.psychsport.2017.03.009.

[11] J. R. Stroop, "Studies of interference in serial verbal reactions," *Journal of Experimental Psychology*, vol. 18, no. 6, pp. 643–662, 1935, doi: 10.1037/h0054651.

[12] M. Biernacki, R. Kennedy, and Ł. Dziuda, "Simulator sickness and its measurement with Simulator Sickness Questionnaire (SSQ)," *Medycyna Pracy*, vol. 67, no. 4, pp. 545–555, Aug. 2016, doi: 10.13075/mp.5893.00512.